COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 | Data Structures and Algorithms	**SECTION:**	CPE21S3
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Seatwork 10.1 **Graph**

1. PROCEDURE OUTPUT

1. Define the following: (10)
    a. **Graph** - A graph is a non-linear data structure made up of nodes or vertices and edges. The edges connect the nodes in the graph.

    b. Vertex - A vertex is a point or element in a graph that represents a node, it is also a point in a graph that edges connect.

    c. Nodes - Nodes are the elements or points in the graph and they can also be called vertices

    d. Weight - Weight is the value associated with an edge in a weighted graph, it represents the cost, distance, time or another value such as W: E -> R.

    e. Path - Path is a sequence of vertices connected by edges from one vertex to another.

    f. Directed graph - Directed graph is a graph where the edges have a specific direction on where they are going.

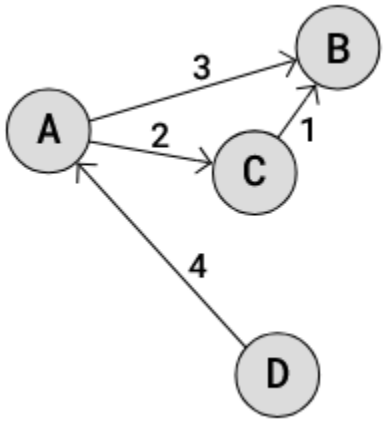
    g. Connected graph - In an undirected graph, a connected component is a subset of vertices that are all reachable from each other. The graph is connected if it contains exactly one connected component, meaning every vertex is reachable from every other. In a directed graph, a strongly connected component is a subset of mutually reachable vertices, meaning there is a path between any two vertices in the set.

    h. Directed cyclic - Directed cyclic graph also known as DAG is a graph that contains a cycle, where the path will return to the starting point eventually.

    i. Adjacency matrix - An adjacency matrix is a 2D matrix that is being used to represent a graph, such as matrix[i][j] is equal to 1 indicates that an edge from node i to node j.

j. **Adjacency List -** An adjacency list uses an array of lists where each list contains the nodes adjacent to a specific node.

2. Identify the parts of the following graph. (10)



- **Type of graph** - This is a directed weighted graph
- **Vertices** - A, B, C, D
- **Edges**
  - (A -> B) with a weight of 3
  - (A -> C) with a weight of 2
  - (C -> B) with a weight of 1
  - (D -> A) with a weight of 4
- **Paths** - here is the example path for the figure above that it can take.
  - **D -> A -> C -> B**, which has a total weight of 7 because 4 + 2 + 1 = 7.

3. Plot the adjacency matrix of the graph above. (10 pts)
The order that we will be using to plot the adjacency matrix is A, B, C, D. Since the graph is a directed weighted graph, w(u,v) is stored at the (u,v) position of the matrix, and if there is no edge between the two vertices we put 0.

	A	B	C	D
A	**0**	**3**	**2**	**0**
B	**0**	**0**	**0**	**0**
C	**0**	**1**	**0**	**0**

D	4	0	0	0

4. Create a program in C++ program to create a graph that looks like in the above figure.

```cpp
#include <iostream>

int main() {
    char letters[4] = {'A', 'B', 'C', 'D'};
    int graph[4][4] = {0}; // creates a graph

    graph[0][1] = 3; // A -> B
    graph[0][2] = 2; // A -> C
    graph[2][1] = 1; // C -> B
    graph[3][0] = 4; // D -> A

    std::cout << "Adjacency Matrix:\n";
    std::cout << "  A B C D" << std::endl;
    for (int i = 0; i < 4; i++) {
        std::cout << letters[i] << " ";
        for (int j = 0; j < 4; j++) {
            std::cout << graph[i][j] << " ";
        }
        std::cout << std::endl;
    }
    return 0;
}
```

```
Adjacency Matrix:
  A B C D
A 0 3 2 0
B 0 0 0 0
C 0 1 0 0
D 4 0 0 0
```

5. Differentiate BFS vs DFS types of search.

- Breadth First Search (BFS) works by traversing the graph layer by layer by visiting the neighboring nodes first of the current note and BFS uses a queue style implementation. Depth First Search (DFS) works by traversing the graph by going down as far as possible on the path before it backtracks on the path and DFS uses a stack style of implementation.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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